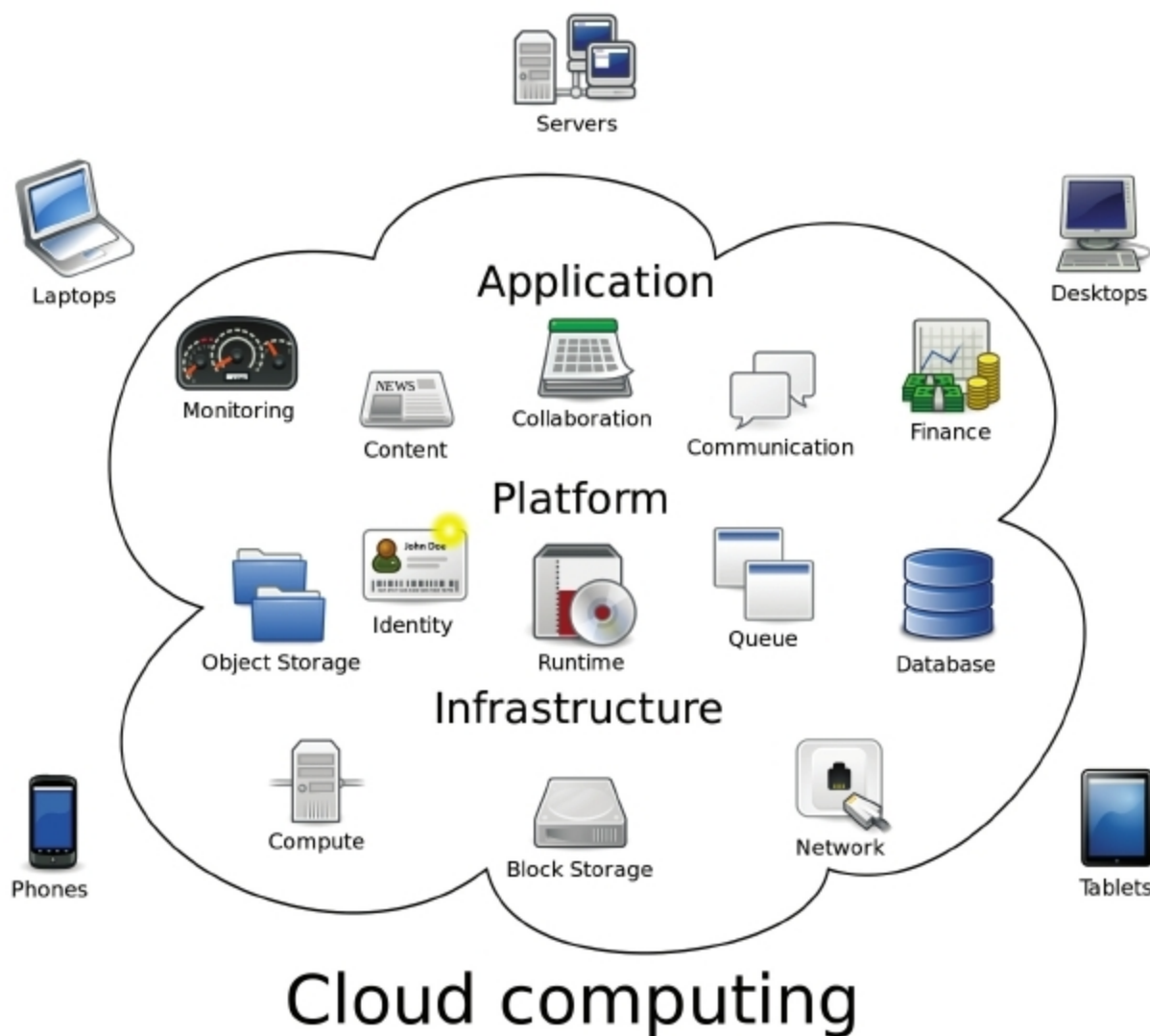


The Benefits and Challenges of the Internet as a Service for Cloud Computing

IaaS provides high performance computing on supercomputers, computer grids or computer clusters. This helps solve complex problems involving millions of variables or calculations. Examples include earthquake and protein folding simulations, climate and weather predictions, financial modelling and product design evaluation.



In an IaaS model, a cloud provider hosts infrastructure components traditionally in a data centre, which includes servers, storage, networking hardware, as well as virtualisation or hypervisor layer.

An IaaS provider also supplies a range of services to companies, including detailed billing, monitoring, log access, security, load balancing, clustering, backup, replication and recovery. IaaS provides greater level of automation and orchestration for important infrastructure tasks.

IaaS customers can access resources and services through WAN, such as the Internet. For example, they can log into the IaaS platform to create virtual machines. They can deploy middleware such as databases, create storage buckets for workloads and backups, and install enterprise workload onto the virtual machines. Users can then use IaaS to track costs, monitor performance, balance network traffic, troubleshoot application issues, manage disaster recovery and so on.

Any cloud computing model requires participation from a provider, who is a third-party organisation specialised in IaaS. Amazon Web Services (AWS) and Google Cloud Platform (GCP) are independent IaaS providers. An enterprise may also opt for private cloud to become its own provider of infrastructure services.

AWS offers storage services such as Simple Storage Services (S3) and Glacier, as well as computer services

Cloud computing is growing rapidly and involves potentially greater exposure to security threats and privacy breaches, especially when it comes to private and public clouds. It is a new way of providing computed resources and applications on-demand. It has become one of the most important research topics in computer science and information systems today.

Three main categories of cloud computing are:

- Software as a Service (SaaS)

- Platform as a Service (PaaS)
- Internet as a Service (IaaS)

A form of cloud computing, IaaS provides virtualised computing resources over the Internet. It is a cloud model where units of computation, in the form of virtual machines and/or storage, are allocated to customers, who then access their assigned resources via wide area network (WAN). Cloud system consumers are granted complete control of all resources assigned to them, but they have no control over the positioning layer.